

DARCIEA ARULNESAN

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Master of Engineering graduate with strong analytical and problem-solving skills gained through technical projects and teaching experience. Seeking a PhD position in the field of Affective Computing and Assistive Robotics.

EDUCATION

Imperial College London, MEng Mechanical Engineering

October 2019 - August 2024

- First Class Honours
- Relevant modules: Advanced Control (81%), Introduction to Robotics (81%), Future Clean Transport Technology (73%), Human Neuromechanical Control and Learning (67%), Machine Learning (66%)
- Successfully completed several team projects, developing products from initial ideation to delivery, demonstrating skills and competencies in project management, teamwork and engineering design.
- Independently compiled multiple lab reports, analysed experimental data, and effectively communicated results, culminating in a Master's dissertation.
- Paused studies for family reasons, pursued diverse skills, clarified career interests, and returned to studies with increased confidence and stronger academic performance.

Wimbledon High School, London

2012 - 2019

- Awarded GDST Lorna Cocking Scholarship and HSBC Scholarship
- A Levels: Maths (A*), Further Maths (A), Physics (A), Chemistry (A)
- GCSES: 10 A*s (including Maths and Science) and an FSMQ (Additional Maths)

ENGINEERING PROJECTS

MNIST Digit Classifier

February 2025 - Present

- Building, containerizing, and deploying a simple digit recognizer trained on the MNIST dataset.
- Trained Pytorch models locally, achieving an accuracy of 98.42% using CNNs.
- Developed a Streamlit web app where users can draw a digit, submit it, and receive a prediction with confidence scores. Users can also provide true labels for data collection.
- Logged the timestamp, predictions, true label from Streamlit web app into a PostgreSQL database.
- **Next steps:** Containerizing the PyTorch model, Streamlit app, and database using Docker, and deploying to a self-managed server with public access.

MEng Individual Project - A Wearable Sensor to Measure Fetal Movements

2023-24

- Clinically validated a novel multi-modal wearable fetal movement monitor developed by the Imperial College Biomechatronics lab, enabling accurate at-home monitoring.
- Conducted high-quality data collection from over 50 pregnant participants, integrating real-time ultrasound observations to enhance dataset reliability.
- Applied supervised learning with a transformer-based model using Tensorflow to perform binary classification of fetal movements, improving distinction and detection accuracy.

Introduction to Robotics Projects

2023-2024

- Designed a 4-DoF robotic manipulator using SolidWorks, performing kinematic analysis, and computational validation in MATLAB.
- Developed a mobile robot to follow a path by extracting spatial map features using computer vision, applying algorithms to process and analyze map data for pathfinding.

Award-Winning 3rd Year Design, Make and Test Project

2022-23

- Awarded Arup Prize for Third Year Group Project
- Designed a menstrual product testing rig to address the lack of safety standards surrounding menstrual products and serve as a new regulated standard.
- Served as the Mechatronics Lead, responsible for designing and implementing the control system to maintain body temperature.
- Produced a Testing and Analysis report, evaluating the rig's performance, and suggesting improvements, showcasing strong analytical, documentation, and communication skills.

PROFESSIONAL DEVELOPMENT

Self-Driving Cars Specialization

September 2023

University of Toronto

- Completed a four course specialisation consisting of: Introduction to Self-Driving Cars, State Estimation and Localization, Visual Perception and Motion Planning.
- Gained practical knowledge of the technologies and algorithms used in autonomous vehicles, with a focus on sensor integration, navigation, and decision-making systems.

Introduction to Programming the Internet of Things Specialization

August 2023

University of California, Irvine

- Gained hands-on experience in C programming with Arduino, Linux systems using Raspberry Pi, and the fundamentals of Internet of Things (IoT) and Embedded Systems.

Wellbeing Rep for the Women in Mechanical Engineering Society

2021 - 2022

- Founded the new Women in Mechanical Engineering Society at Imperial College to address the challenges of gender inequality in STEM.
- Created a supportive community and provided opportunities for female students in a male-dominated field.

WORK EXPERIENCE

Preschool Teacher, Keren's Nursery

2022 - 2024

- Undertook a full-time role during a pause from studies, gaining valuable soft skills through experience in a contrasting employment sector.
- Worked with young children, demanding constant patience, adaptability, and the ability to manage dynamic situations, including ensuring safety in unpredictable scenarios.
- Key skills developed: effective communication, keen attention to detail, teamwork, problem-solving, ethical decision-making.

Physics Teaching Internship, Department for Education

July 2023

- Delivered a careers talk to female students interested in STEM, showcasing university projects and offering insights into pursuing further education in engineering.
- Taught and assisted in Physics lessons, demonstrating competency in explaining complex concepts clearly to students of varying abilities.

Python, Maths, Physics Tutor, FireTech Coding Camp Leader

2022 - 2024

- Led week-long Python coding camps for young teens, introducing them to programming fundamentals and fostering problem-solving skills.
- Regularly tutored students in A-Level and GCSE Maths and Physics.